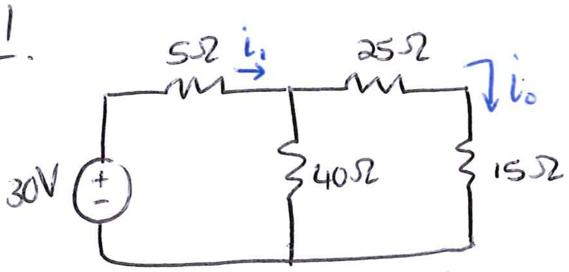
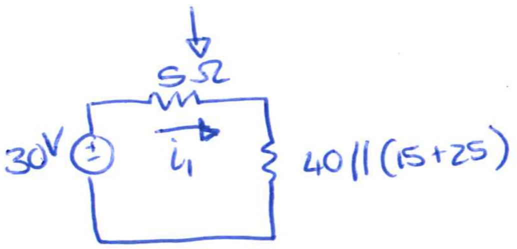
Tutorial 4 - Memo

4.1



- Calculate i_o.
- What value of input voltage will make $i_o = 5A$?



$* \; 40 \,||\, (25+15) = 20\Omega.$

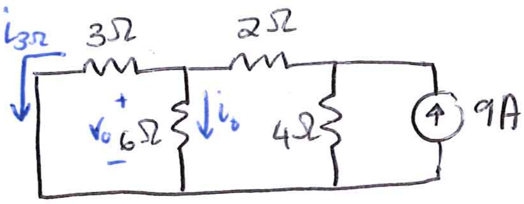
$* \; i = \dfrac{30}{(5+20)} = 1.2A$

$\therefore \; i_o = i\left(\dfrac{40}{40+40}\right) = 0,6A.$

$\therefore$ Use linearity to determine V_{in} for $i_o = 5A$.

$$\Rightarrow \begin{array}{l} 0,6 \; R = 30 \\ 5 \; R = V_{in} \end{array} \Rightarrow V_{in} = \frac{30 \times 5}{0,6} = 250V$$

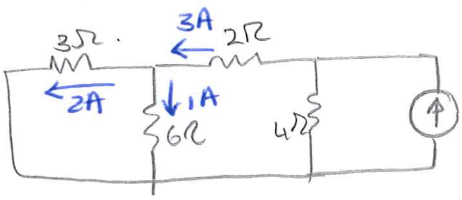
4.4



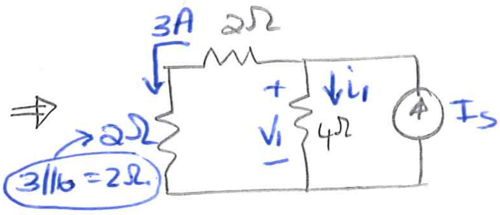
Use linearity to determine i_o.

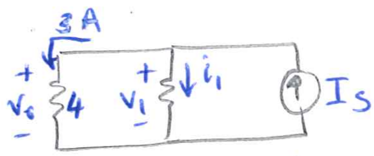
$*$ If $i_o = 1A$; then $V_o = 6V$ and $i_{3\Omega} = 2A$

$\therefore$



$\Rightarrow$





$\therefore V_o = 3(4) = 12V$

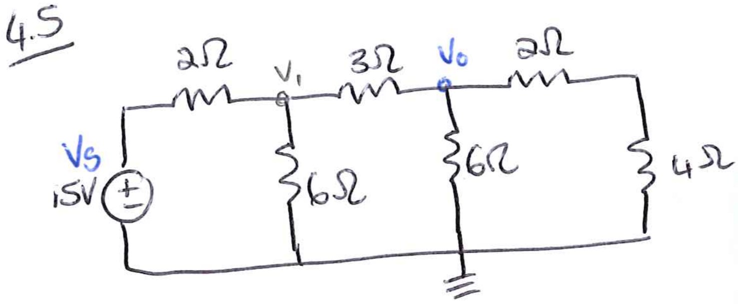
$\Rightarrow I_s = 3+3$

Hence, if:

$I_s = 6A \rightarrow I_o = 1A$

$I_s = 9A \rightarrow I_o = \frac{9}{6} = 1,5A.$

4.5



Assume $V_o = 1V$ and use linearity to find actual value for V_o.

→ If $V_o = 1V$:

① $\frac{V_o}{6} + \frac{V_o}{6} + \frac{V_o - V_1}{3} = 0$

$2V_o + 2V_o - 2V_1 = 0$

$2V_1 = 4(1)$

$V_1 = 2V$

② ∴ $\frac{V_1 - V_s}{2} + \frac{V_1}{6} + \frac{V_1 - V_o}{3} = 0$

$3(2) - 3V_s + (2) + 2(2) - 2(1) = 0$

$10 = 3V_s$

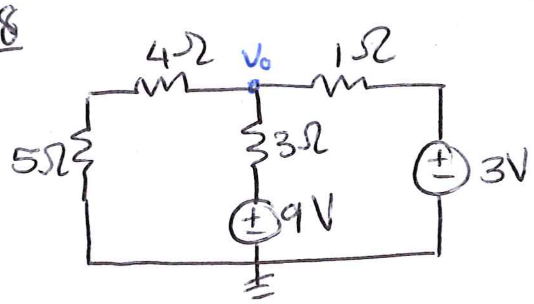
$V_s = \frac{10}{3}$

⇒ Thus,

$V_s = \frac{10}{3} \rightarrow V_o = 1$

$V_s = 15 \rightarrow V_o = \frac{15}{(10/3)} = 4{,}5\ V$

4.8

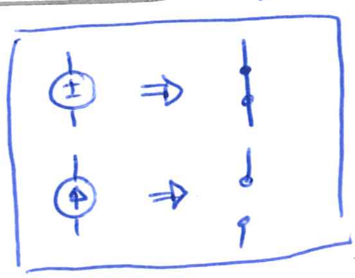


Use SUPERPOSITION to find V_o.

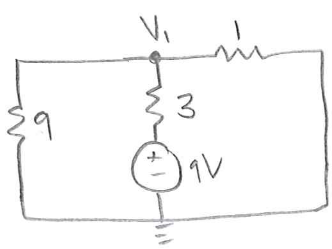
* Independent sources *

(V_1 and V_2 are due to 3V- and 9V-sources respectively)

⇒ $\boxed{V_o = V_1 + V_2}$



For V_1:



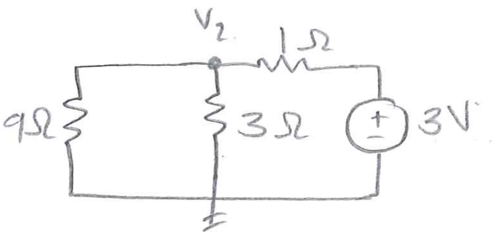
$\frac{V_1}{9} + \frac{V_1 - 9}{3} + \frac{V_1}{1} = 0$

$V_1 + 3V_1 - 27 + 9V_1 = 0$

$13V_1 = 27$

$V_1 = 2{,}077\ V$

For V_2 :



$$\frac{V_2}{9} + \frac{V_2}{3} + \frac{V_2 - 3}{1} = 0$$

$$V_2 + 3V_2 + 9V_2 = 27$$

$$13V_2 = 27$$

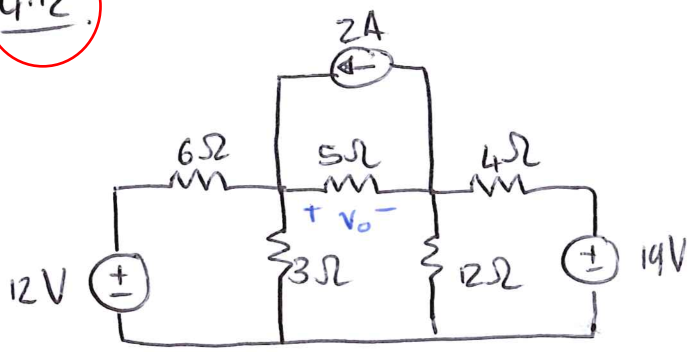
$$V_2 = 2{,}077 \text{ V.}$$

$$\therefore V_o = V_1 + V_2$$

$$= 2{,}077 + 2{,}077$$

$$= 4{,}15 \text{ V}$$

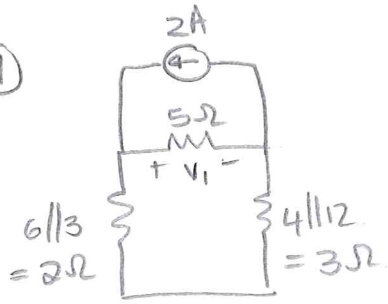
4.12.



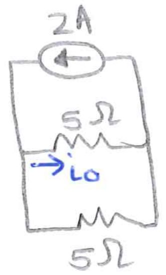
Determine V_o using SUPERPOSITION.

[* 3 independent sources
$\Rightarrow V_o = V_1 + V_2 + V_3$]

①



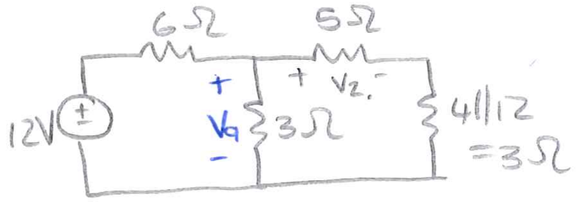
$\Rightarrow$



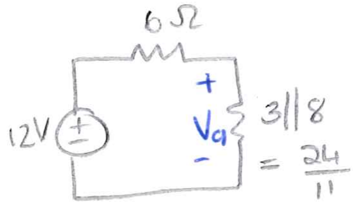
$$i_o = 2\left(\frac{5}{5+5}\right) = 1 \text{ A}$$

$$\therefore V_1 = 5(1) = 5 \text{ V}$$

②



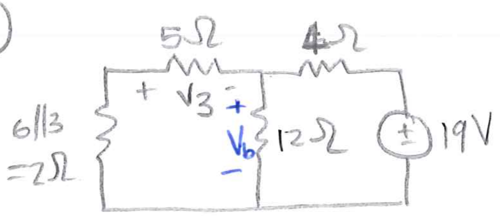
$\Rightarrow$



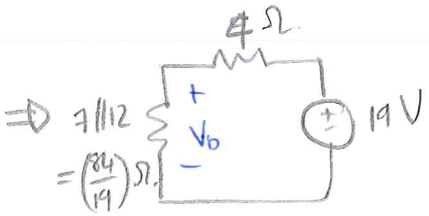
$$* \ V_a = 12\left(\frac{24/11}{6 + 24/11}\right) = 3{,}2.$$

$$\therefore V_2 = V_a\left(\frac{5}{5+3}\right) = 2 \text{ V}$$

③



$\Rightarrow$



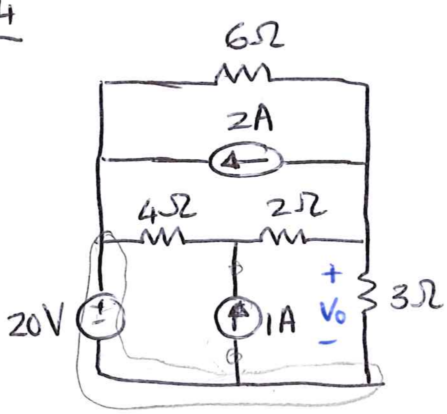
$$V_b = 19\left(\frac{84/19}{84/19+4}\right) = 9.975$$

$$\therefore V_3 = V_b\left(\frac{-5}{5+2}\right)$$

$$= -7{,}125$$

$$\therefore V_0 = 5 + 2 - 7{,}125 = -0{,}125\ V$$

4.14

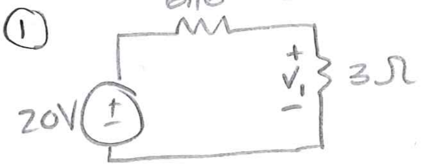


Use SUPERPOSITION to determine V_0

$$\boxed{V_0 = V_1 + V_2 + V_3}$$

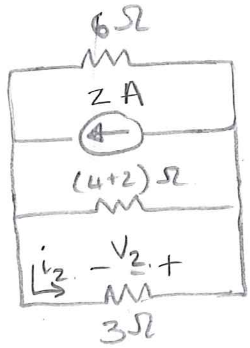
①

6//6 = 3Ω



$$\Rightarrow V_1 = 20\left(\frac{3}{3+3}\right) = 10\ V$$

②



$\Rightarrow$

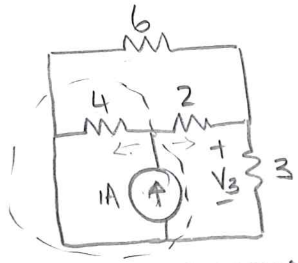
6//3 = 2Ω.

$$* \ i_q = 2\left(\frac{6}{6+2}\right) = 1{,}5\ A.$$

$$\therefore i_2 = 1{,}5\left(\frac{6}{6+3}\right) = 1\ A.$$

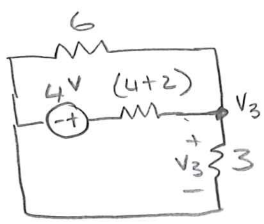
$$\therefore V_2 = 3(-1) = -3\ V$$

③



SOURCE TRANSFORMATION

$\Rightarrow$



$$\frac{V_3}{3} + \frac{V_3 - 4}{6} + \frac{V_3}{6} = 0$$

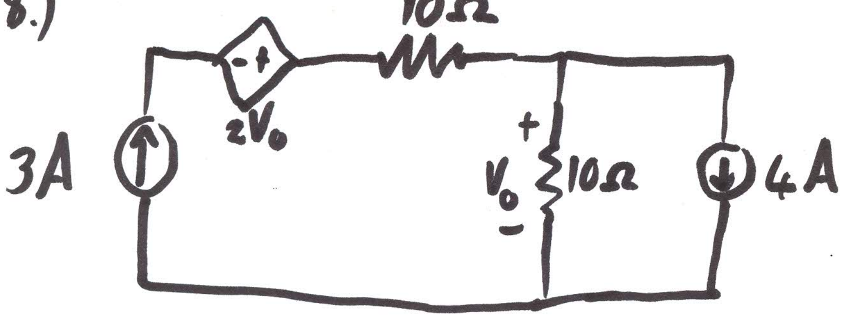
$$2V_3 + V_3 - 4 + V_3 = 0$$

$$4V_3 = 4.$$

$$V_3 = 1\ V.$$

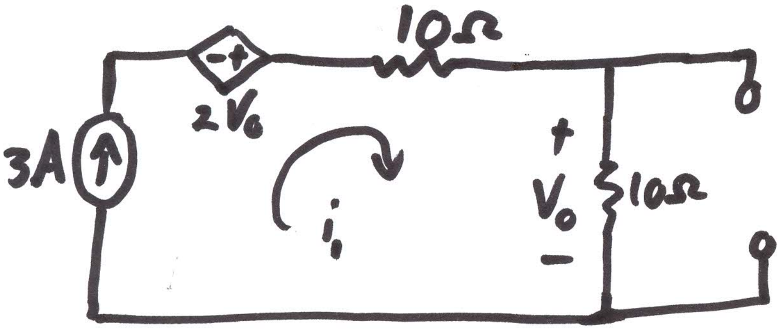
$$\Rightarrow V_0 = 10 - 3 + 1 = 8V$$

4.18.)



Use superposition to find V_o.

① For only the 3A current source:



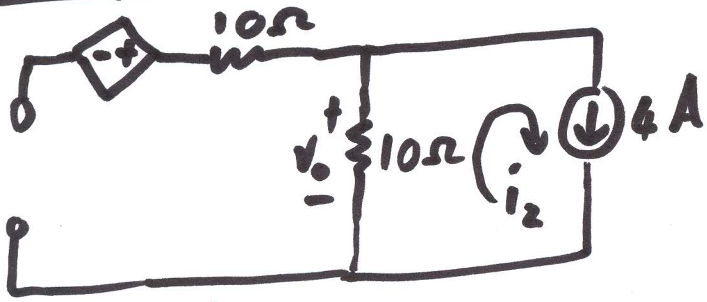
$i_1 = 3A$

$\therefore V_o = i_1(10)$

$V_o = (3)(10)$

$\underline{V_o = 30V}$

② For only the 4A current source:



$i_2 = 4A$

$-V_o = i_2(10)$

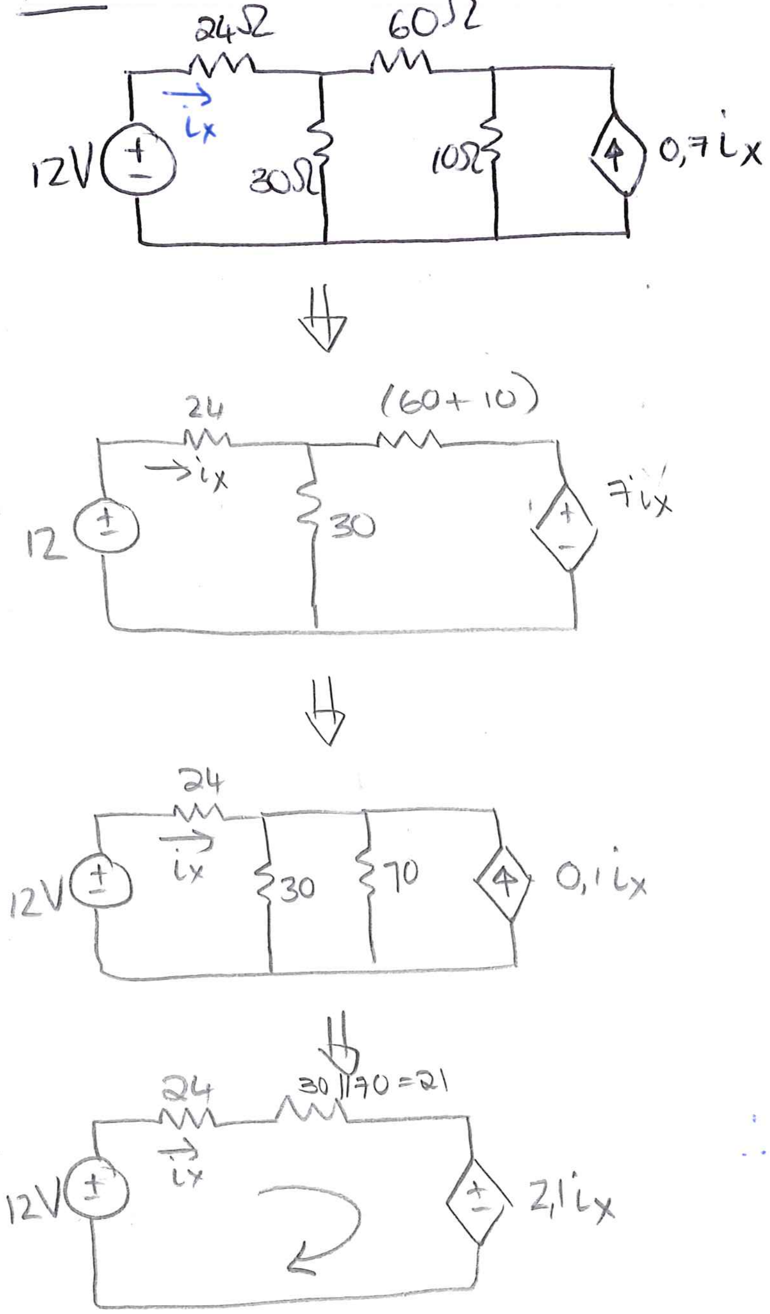
$\underline{V_o = -40V}$

③ $V_o = V_{o_1} + V_{o_2}$

$V_o = 30 + (-40)$

$\underline{V_o = -10V}$

4.30



Find i_x using Source transformation.

$$\therefore \; -12 + 24i_x + 21i_x + 2{,}1i_x = 0$$

$$47{,}1\, i_x = 12$$

$$i_x = 0{,}25 \text{ A}$$

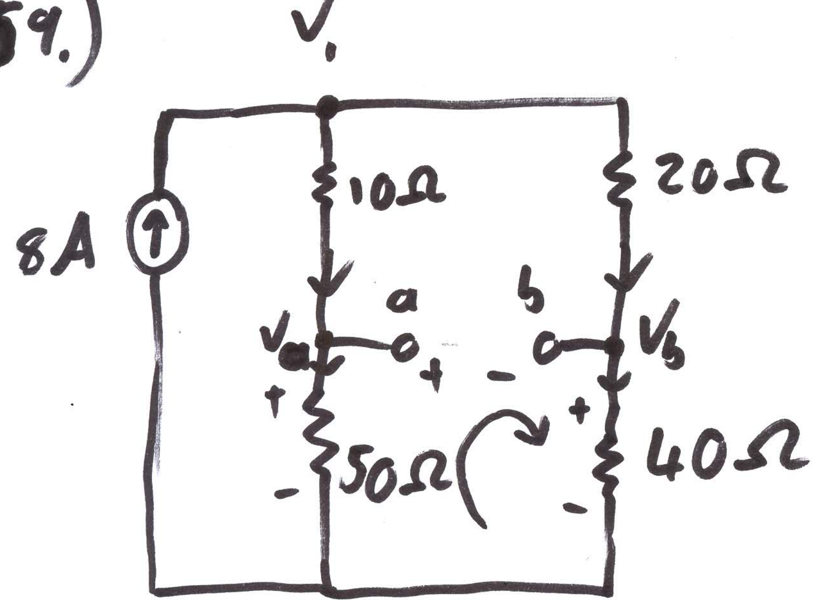
⑤ KVL i_2: $(V_{ab} = V_{th})$

$$-V_2 + (V_2 - V_3) + V_{th} = 0$$

$$-(-4) + 80 + V_{th} = 0$$

$$V_{th} = -84V$$

4.59.)

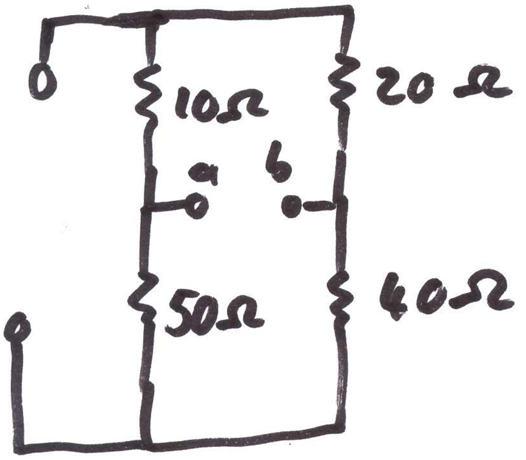


$R_{th} = ?$

$V_{th} = ?$

$I_N = ?$

R_{th}:



$$R_{th} = (10+20) \,||\, (50+40)$$
$$= 30 \,||\, 90$$
$$= \underline{\underline{22{,}5\,\Omega}}$$

V_{th}:

KCL at node V_1:

$$8 = \frac{V_1}{(10+50)} + \frac{V_1}{(20+40)}$$

$$8 = \frac{V_1}{60} + \frac{V_1}{60}$$

$$8 = \frac{V_1}{30}$$

$$\underline{V_1 = 240\,V}$$

Voltage division:

$$V_a = (240)\left(\frac{50}{10+50}\right)$$

$$\underline{V_a = 200\,V}$$

$$V_b = (240)\left(\frac{40}{20+40}\right)$$

$$\underline{V_b = 160\,V}$$

KVL :

$$-V_a + V_{ab} + V_b = 0$$

$$V_{ab} = V_a - V_b$$

$$V_{ab} = 200 - 160$$

$$V_{ab} = 40V$$

$$\therefore \underline{V_{th} = V_{ab} = 40V}$$

$$I_N = \frac{V_{th}}{R_{th}}$$

$$= \frac{40}{22,5}$$

$$= \underline{1,78\,A}$$

4.20 V = 22V

R = 4 Ohm

4.24 Vx = 2.978V

4.39 Rth = 20 Ohm

Vth = -84V

4.40 Rth = 2857.14 Ohm

Vth = 60V